CS 2100: Program Design and Implementation 1

Not an official Northeastern University syllabus. Course code, title and credits come from the institution's public catalog; everything else below is invented.

Credits	5 semester credits	**Term**	Fall 2024
Contact hours	75 hours (lecture + recitation)	**Instructor**	Synthetic Instructor 3
Prerequisites	One introductory course in the discipline	**Office hours**	(not applicable - test file)

Course description

Program Design and Implementation 1. Topics include discrete probability, algorithm efficiency, language paradigms, object-oriented programming and stacks and queues; later weeks address binary trees, sorting algorithms and cryptography basics.

Learning outcomes

By the end of the course, students will be able to:
1. Explain the core ideas of discrete probability.
2. Apply algorithm efficiency to solve standard problems.
3. Analyze problems involving language paradigms.
4. Use object-oriented programming in practical exercises.
5. Evaluate approaches to stacks and queues.
6. Communicate findings about binary trees.

Weekly schedule

Week	Topic	Activity / assessment
1	Discrete probability	Problem set
2	Discrete probability (continued)	In-class exercise
3	Algorithm efficiency	Quiz
4	Language paradigms	Case study
5	Language paradigms (continued)	Problem set
6	Object-oriented programming	Reading and discussion
7	Review and midterm	Midterm exam
8	Stacks and queues	In-class exercise
9	Stacks and queues (continued)	Quiz
10	Binary trees	Case study
11	Sorting algorithms	Problem set
12	Sorting algorithms (continued)	Reading and discussion
13	Cryptography basics	Problem set
14	Review and synthesis	Final exam / project

Assessment

Component	Weight
Problem sets / assignments	25%
Quizzes	15%
Midterm exam	25%
Final exam or project	30%
Participation	5%

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Materials and policies

Required materials: instructor-provided open course notes and readings posted on the course site. Grading scale: A 93-100, A- 90-92, B+ 87-89, B 83-86, B- 80-82, C+ 77-79, C 73-76, C- 70-72, D 60-69, F below 60. Policies: see the institution's own catalog; none are reproduced here.

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